

THE PARIKSHA PROTOCOL

Verifiable Custody and Conduct for Examinations

Specification · Examination Core Profile · draft-pariksha-core-01

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Companion documents: Pramāṇa (policy programme) · On the Truth of Custody, CC7 (kernel theory)

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1. Scope, Terminology, and Conformance Language

This document specifies the Pariksha Protocol Examination Core Profile: the minimal normative core for conducting a public examination such that every artifact is a verifiable credential, every act is bound to an identified participant under explicit authority, and every event is witnessed on a tamper-evident ledger. The profile covers one examination instance from definition to independently verifiable results, over pen-and-paper or device delivery, online or offline.

The key words MUST, MUST NOT, SHOULD, SHOULD NOT, and MAY are to be interpreted as in RFC 2119. “The Protocol” refers to this specification; “an Implementation” refers to any conforming software; the Protocol names none. The Protocol is delivery-agnostic: nothing in the normative core assumes paper, screen, or any particular rendering of the examination content. It is also a two-layer standard: this core, which names no country, identity system, or role; and profiles, which fill those in for one setting. This draft ships with one profile, the Examination Profile — India (Appendix A). A conforming Implementation claims the core plus a named profile.

Design lineage: the Protocol instantiates participant-centric computation — the WHO as a first-class dimension of every event — as developed in the Contextual Compute research programme (see the companion paper, On the Truth of Custody). Familiarity with either is not required to implement the Protocol.

2. The Kernel

The Protocol is four nouns, three verbs, and one semantic condition. Everything else in this document is enumeration and constraint.

2.1 Nouns

PARTICIPANT — any actor able to take a protocol action: human, organisation, or machine. Every Participant is registered (§3) and holds at least one Credential. There are no unregistered actors on the acting side of the Protocol; the sole unregistered role is the Verifier of outcomes (§3.4).

CREDENTIAL — the only artifact type. Roles, papers, responses, grants, and results are all Credentials: signed objects of the form issuer → subject, with claims, an optional encrypted payload, and a proof. The taxonomy is enumerated in §5.

METADATA — the contextual rules of a relationship, carried inside the Credential's claims: whom it is for, when it may open, under what quorum, with what variances. Policy travels inside the artifact it governs; there is no out-of-band rulebook to consult at verification time.

SOCIAL PROOF — verifiable attestation of identity and standing. Identity answers “is this a real, unique person or device”; standing answers “what is this Participant currently authorised to be.” Proof requirements are tiered by the authority a role carries (§4).

2.2 Verbs

ISSUE — the creation ceremony of a Credential: claims composed, payload sealed if present, keys split if quorum-held, proof signed, issuance witnessed.

CUSTODY — the movement and holding of a Credential between issuance and presentation. Custody confers possession, never access: a custodian of a sealed payload holds ciphertext and a witnessed obligation, nothing more.

PRESENT — the consumption ceremony: verification of the Credential and of the presenting Participants' proofs, followed — only on success — by unlock of any sealed payload. Presentation is verify-then-decrypt as a single ceremony; a Credential that fails verification MUST NOT reach decryption.

2.3 The semantic condition: witnessing

Witnessing is not a component; it is the condition under which the verbs count as having happened. Every ISSUE, every CUSTODY transfer, and every PRESENT — including every refused attempt — MUST produce a signed event on the ledger (§8). An action without a witnessed event has not occurred, in the Protocol's semantics. Denials are first-class events with the same evidentiary weight as grants.

3. Participants and the Registry

3.1 Registry rules

R1. Every Participant MUST be registered before acting; registration and revocation are themselves witnessed events. The registry has no silent edits. R2. Roles bind to individuals per proceeding instance and context (“quorum member for instance X at context C”), never globally; scope lives in the Credential. R3. Machines that act — servers, terminals, unlock devices, agents — are Participants with Device Credentials; there are no anonymous system events. R4. Quorum roles MUST span at least two issuing authorities (separation of masters): a body cannot form a quorum entirely from Credentials chaining to its own root.

3.2 Actor taxonomy (profile-defined)

The core does not name roles. A conforming profile MUST define a role taxonomy covering at least: an Authority domain (the root and its policy administrators), an Artifact domain (those who create and seal content), a Custody domain, a Context domain (the roles that form quorums at the place of conduct, plus the validator and terminal devices), a Subject domain (the participants whose sessions are bound), a Machine domain (agents and platform services acting under service credentials), and an Oversight domain (live monitoring and post-hoc audit). Every quorum role MUST be a role the profile names, and the profile MUST state which roles satisfy R4. The taxonomy for Indian public examinations is in Appendix A.

3.3 Contextual locality

A human Participant occupies exactly one physical context at a time. The Protocol treats this as load-bearing: a Candidate Credential **MUST** bind to at most one active session in one context; a second concurrent binding attempt is semantically invalid and **MUST** be rejected and witnessed as a denial. Impersonation-by-parallel-presence is thereby unexpressible rather than merely detectable.

3.4 The one unregistered role

Verification of outcome Credentials (§5, type 10) **MUST** be possible by any party, anywhere, with no registration, account, or permission. The registry's boundary is meaningful precisely because outcome trust is consumed permissionlessly while all authority is exercised accountably.

4. Social Proof

4.1 The identity floor

For human Participants, the identity layer **MUST** anchor to the identity substrate the profile names — a national or institutional identity system capable of unique-person assurance — captured at Credential issuance and sealed into the Credential's `identity_binding` claim. Presentation consumes this proof locally (live biometric or equivalent against the sealed template) and **MUST NOT** require network access to the identity substrate at presentation time. For device Participants, the identity layer is hardware attestation (attested boot, certified capability set). A profile **MUST** state its identity substrate and **MUST** provide an alternate identity-proof path for legitimate exceptions the substrate does not cover. India's binding is in Appendix A.

4.2 Standing, layered above identity

Standing — current authorisation in a role — chains above identity as separately revocable claims: identity does not revoke, roles do. Both layers ride one Credential and verify in one presentation. In v1 presentations are plain signed disclosures; from v1.1 the `presentation_proof` field carries a zero-knowledge proof of the conjunction (“identity-verified AND valid standing for this context”) without disclosing identity to the ledger.

4.3 Proof tiers

Proof accumulates with authority. The `required_proof_level` claim in every Role Credential is enforced by the grant engine: presentation below tier is a denial.

Tier	Proof required	Authority carried
1	Identity floor (§4.1)	power over one's own session only
2	Tier 1 + institutional attestation, per instance	operational roles: handling, invigilating, monitoring
3	Tier 2 + cross-authority endorsement (attester outside the operating body's root)	quorum roles; sealing roles

Tier	Proof required	Authority carried
4	Tier 3 + multi-party issuance ceremony; high-consequence acts require fresh quorum co-signature at time of act	roles that touch content or rules; the Authority root itself

A profile MUST assign each of its named roles to a tier. The assignment for Indian public examinations is in Appendix A.

5. The Credential Taxonomy

Twelve types, four families, one shape: issuer → subject, claims, optional payload, proof. Cross-cutting requirements: every Credential MUST carry scope (context-bound, never global), issuing_authority, revocation_registry_uri, and presentation_proof_type (plain in v1; zk from v1.1). Policy variance from a baseline MUST be drawn from an enumerated set and MUST reference a witnessed justification event (bounded degrees of freedom: window width, quorum size, observer count).

Family 1 — Identity and role

#	Credential	Issuer → Subject	Key metadata claims
1	Authority	root ceremony → examination body	jurisdiction; exam_types; validity; revocation_registry_uri
2	Role	authority/registrar → official	role; scope (exam-instance + context); identity_binding (sealed); issuing_authority; required_proof_level; validity_window
3	Device	certification → device	device_class (validator terminal render quorum-app); attestation; certified_capabilities; context_binding (per instance); validity (per exam-day)
4	Candidate	authority → candidate	candidate_id (profile-defined); identity_binding (face template, sealed to centre); registered_exam; assigned_context (one hall); accommodations (incl. nominee ref)

Family 2 — Artifact

#	Credential	Issuer → Subject	Key metadata claims
5	Exam Paper	setter/assembly authority → receiving context (centre; candidate at later rungs)	exam_id; centre_binding; unlock_window; quorum_policy; payload_sha256; watermark_commitment; language; form_id; policy_variance_ref; payload = P2P-encrypted paper; keyShares[] per quorum member
6	Response	bound session (device + candidate) → evaluation authority	session_ref; candidate_ref; exam_id; chain_head; response_count; timestamps; receipt_code; payload = responses sealed to collection key
7	Marks-List	controller → exam instance	exam_id; evaluation_method; result_set_hash; normalization_ref (published equating parameters); ledger_anchor; signed_ts

Family 3 — Authority in motion

#	Credential	Issuer → Subject	Key metadata claims
8	Grant	grant engine → requesting participant(s)	scope (e.g. decrypt:PKG-C012); quorum_satisfied (co-signing credentials); window; expires_on_use = true; context
9	Capability Token	grant engine at last sync → centre context	as Grant, plus offline_valid = true; sync_deadline; local_chain_required = true

Family 4 — Outcome

#	Credential	Issuer → Subject	Key metadata claims
10	Result	authority (via marks-list chain) → candidate / candidate_id	exam_id; score; percentile; rank; marks_list_ref; ledger_anchor; revocation_status_uri; issued_ts — verifiable by anyone, no registration
11	Admit	authority → candidate	exam_id; centre; seat; report_time; identity_binding_ref (into #4); single_use_binding = true (consumed at session bind)
12	Integrity Certificate	authority + independent observer authority, co-signed → exam instance	exam_id; centres_reporting; chain_root (Merkle root over centre subtrees); anomalies {raised, resolved}; unbroken

Structural symmetries an Implementation SHOULD preserve: type 6 is type 5 in reverse (paper in, responses out, same custody discipline); types 8 and 9 differ by one boolean; type 11 consumes itself into a session binding; type 12 is the federation checkpoint (§6.3) wearing a signature.

6. Contexts and Nodes

6.1 The context

A context is a bounded interaction locale with its own participant set and event stream — in the Core Profile, one examination centre for one exam instance. Contexts do not contend: no event in one centre's stream conflicts with another's, so no global consensus is required during conduct. Consensus is needed only where contexts touch — at federation checkpoints.

6.2 The context node (centre validator)

Each centre runs a validator: a Device Credential holder (class validator) granted, per exam instance, the scoped authority validator:centre:exam>window. The validator verifies presentations locally (signature verification requires no network), anchors the centre's hash chain, arbitrates device sessions over LAN, and witnesses all centre events in real time. Validator authority is just-in-time and expiring like all authority in the Protocol: the physical machine MAY differ every day; the role is granted fresh through the standard ceremony. Any number of validators MAY serve one context for redundancy; they share one chain.

6.3 Federation

On connectivity, each context checkpoints: its chain head, countersigned by the centre quorum, merges into the national ledger as a subtree. The national chain is a Merkle forest of context subtrees under one root; the root, co-signed, is the Integrity Certificate (#12). Tampering within any context breaks that context's subtree visibly and is contained to it — blast radius one centre, at the infrastructure layer.

7. Processes

7.1 ISSUE

Credential issuance **MUST** be a witnessed ceremony appropriate to the type's proof tier. For the Exam Paper Credential specifically: (a) content is finalised; (b) per-receiver watermark is embedded; (c) payload is encrypted per receiving context; (d) the content key is threshold-split (e.g. Shamir 2-of-3) and shares wrapped to each quorum member's credential key; (e) claims are composed (window, quorum policy, hashes, variance refs); (f) the assembler signs; (g) plaintext is destroyed, and the destruction is attested by the sealing witness as its own event. Steps (a)-(g) each emit witnessed events; `papervc.issued` and `plaintext.destroyed` are mandatory.

7.2 CUSTODY

Every transfer of a sealed Credential between Participants **MUST** emit `custody.transfer` naming both parties' credentials. Pre-staging sealed Credentials at their destination context ahead of time is **RECOMMENDED** (connectivity resilience): custody of ciphertext carries no access risk by construction. Storage between transfers binds to a custodian credential. There are no anonymous hops: a Credential whose custody chain has a gap **MUST** fail verification at presentation.

7.3 PRESENT

Presentation is one ceremony in two mandatory stages. **VERIFY**: the presenting context validates the Credential's proof (issuer signature), its claims against the moment (window open, context matches, revocations clear), and the presenting Participants' social proof at the required tier. Any failure is a witnessed denial; the ceremony ends. **UNLOCK**: only after full verification, quorum members' shares unwrap and the content key reconstructs inside the presenting device's secure enclave; payload decrypts; the key is zeroised and the grant expires on use. For sessions (device delivery): session binding presents the Admit Credential plus live biometric match against the sealed template, locally; the bound session then signs each response into the context chain, and session completion issues the Response Credential with the candidate's `receipt_code` displayed.

8. Witnessing and the Event

Every event is an instance of one schema (below). Normative rules: E1 — participant is mandatory; there are no system events without a credential. E2 — events hash-chain per context (`prev`) and anchor to the ledger. E3 — denials are events, not log lines; a refused presentation carries the refusing rule and the presenting credential. E4 — policy changes, registrations, and revocations are events. E5 — the ledger is permissioned with encrypted participant fields in v1; it **MUST NOT** function as a public directory of officials. `presentation_proof` is null in v1 and carries the zero-knowledge proof from v1.1.

```
{  
  "event_id": "evt_01J9...",
```

```

"type": "key.released",          // or: papercv.issued, custody.transfer,
                                //    papercv.verified, artifact.decrypted,
                                //    session.bound, response.recorded,
                                //    session.completed, denial.*, ...

"exam_id": "BOARD-2026-P1",
"context": "centre:C012",
"artifact": { "credential_id": "PKG-C012", "sha256": "9f3a..." },
"participant": { "credential": "vc:example:off:8842",
                  "role": "centre.super" }, // role names are profile-defined
"grant": { "grant_id": "grt_7be1", "scope": "decrypt:PKG-C012",
            "quorum": "2-of-3", "window": { "start": "...", "end": "..." } },
"witness": { "ledger_anchor": "ledger:tx:4c19...", "ts": "..." },
"presentation_proof": null,
"prev": "sha256:1d0c..."
}

```

9. Offline Operation

Offline is a first-class mode, not degradation. The invariant: everything that requires the network happens before exam day; exam day verifies signatures and matches biometrics locally. During the last connected sync a context receives its sealed Paper Credentials, its roster of Admit/Candidate Credentials with sealed biometric templates, and Capability Tokens pre-authorising the window's grants. Offline, the validator verifies and grants locally; all events chain into the sealed local log. On reconnection the buffered subtree checkpoints (§6.3); an unbroken chain proves nothing was inserted, altered, or reordered while dark. Submission and transmission are distinct: a Response Credential is complete when its chain closes on-device, regardless of when it uploads. Cutting a context's connectivity therefore gains an attacker nothing and costs the examination nothing.

10. The Physical Bridge

Where delivery is paper, plaintext exists between decrypt and collection. The Protocol does not claim this window away; it compresses and instruments it. Requirements: render **MUST** occur on a certified device credentialed for the render capability, directly to the output path (no export, no file at rest); the render event **MUST** record page and copy counts, and collection **MUST** reconcile against those witnessed counts; the watermark — embedded before sealing — carries provenance across the barrier, so any photograph or copy of rendered output attributes its context; and the plaintext interval **MUST** be bracketed by room.sealed and distribution.completed events attested by the observer. The unlock window is per-context metadata sized to rendering logistics (duplicator-class printing for short papers; wider windows or district render hubs for booklet-scale exams; none of this changes the Protocol — only claims). Device delivery retires the bridge entirely: decrypt-in-app, render-on-screen, zeroise-on-submit.

11. Security Considerations

S1 — The chain authorises and witnesses; the enclave decrypts. No decryption key or share ever touches the ledger; reconstruction occurs only in secure enclaves and is zeroised on use. S2 — Audited primitives only; no novel cryptography anywhere in a conforming Implementation. S3 — Separation of masters (§3.1 R4) is structural, not procedural: a quorum whose proofs chain to one root MUST fail verification. S4 — Bounded variance: per-receiver policy freedom is drawn from enumerated sets, and each variance references a witnessed justification; a compromised issuer cannot quietly weaken one context's rules. S5 — Privacy: the ledger is not a directory (§8 E5); v1.1 zero-knowledge presentations remove identity from the ledger entirely. S6 — Honest residue: within the plaintext window, capture by a present, credentialed insider is not preventable — it is compressed (minutes), attributed (watermark + named custody), and contained (per-context artifacts at later rungs). The Protocol's guarantee is that theft outside the witnessed, quorum-held window is cryptographically meaningless, and inside it is named, timed, and localised.

12. Core Profile Conformance Checklist

An Implementation conforms to the Core Profile if and only if it implements: the registry with rules R1-R4; social-proof tiers 1-3 with the identity floor and device attestation of §4.1 (tier 4 ceremonies MAY be simulated); Credential types 1-6, 8-12 with the cross-cutting claims (type 7 MAY be folded into 12 for a single-examination deployment); the three process ceremonies of §7 with verify-then-decrypt ordering enforced; the event schema of §8 with rules E1-E5; one context node operating fully offline through a complete conduct cycle with successful federation on reconnect; the physical-bridge requirements of §10 for at least one paper-delivered context and, RECOMMENDED, one device-delivered session demonstrating the bridge's retirement; and security considerations S1-S5 without exception. An Implementation MUST additionally declare the profile it conforms to; conformance is always stated as core plus a named profile (e.g. “Pariksha Core, Examination Profile — India, draft-01”).

Everything beyond this checklist — multi-form assembly, item banks, equating, per-candidate delivery at scale, agentic orchestration — is later rungs of the same kernel: new claims and new receivers, no new nouns, no new verbs.

Appendix A — Examination Profile: India

Profile identifier: pariksha:profile:exam-in:draft-01. This profile is normative. It fills the blanks the core leaves open — identity substrate, candidate identifier, role taxonomy, tier assignment, quorum composition — for public examinations in India. An Implementation claiming this profile MUST meet everything below in addition to the core conformance checklist (§12).

A.1 Identity floor

Human Participants: Aadhaar-verified via DigiLocker (including UIDAI offline artifacts), captured at Credential issuance and sealed into identity_binding. Exam-day verification is local against the sealed template; no network call to UIDAI at presentation. Device Participants: hardware attestation (attested boot, certified capability set). Candidate identifier: APAAR. Wallet and verification endpoint for outcome Credentials: DigiLocker. Where a legally recognised signature is required on authoring, review, or custody events: eSign. Consent for sharing outcome data beyond the admission process: a DEPA-pattern consent artefact, expressed as claims. Implementations MUST support an alternate identity-proof path for legitimate exceptions (e.g. non-resident candidates); its specification is deferred to a later draft of this profile.

A.2 Actor taxonomy

Domain	Roles
Authority	examination body (root); board.controller; policy administrator
Artifact	paper setter; paper assembler (issuer of the Paper Credential); sealing witness; translator/moderator
Custody	custody.courier; strongroom custodian
Centre	centre.super, centre.observer, centre.boardrep (quorum); invigilator; centre validator device; render/terminal devices
Candidate	candidate; nominee/scribe (witnessed delegation)
Machine	assembly, custody, and integrity agents (later rungs); platform services (grant engine, ledger) under service credentials
Oversight	board.duty (console, read+acknowledge); auditor (post-hoc, scoped read)

A.3 Proof tiers (assignment of roles)

Tier	Proof	Roles
1	Aadhaar identity	candidate
2	Tier 1 + institutional attestation (employer/board, per instance)	invigilator, courier, custodian, duty officer
3	Tier 2 + cross-authority endorsement (attester outside the board's root)	quorum roles: centre.super, centre.observer, centre.boardrep; sealing officer
4	Tier 3 + multi-party issuance ceremony; fresh quorum co-signature at time of act	setter, assembler, controller, policy administrator; the Authority root itself

A.4 Separation of masters

Registry rule R4 applies: a centre quorum MUST include at least one Credential chaining to a root other than the conducting body's — in practice the independent observer. A quorum whose proofs all chain to one root MUST fail verification (S3). The quorum policy for this profile is 2-of-3 by default, 3-of-3 by policy.

A.5 Physical bridge parameters

The unlock window is per-context metadata sized to rendering logistics: duplicator-class printing for short papers; wider windows or district render hubs for booklet-scale examinations. The core does not change; only the claims do.